

Class 8 — Monitoring Tools

Lesson 3.1 Categories of monitoring tools

Lesson 3.2 Selecting the right tool for a given need

Saturday, September 26, 2026 · 2 p.m. – 6 p.m. · online

Topic 3 · 60-minute lecture · Instructor: Asen Grozdanov

Quiz 2 (5 %, all material to date) is written at the end of this class — separate document



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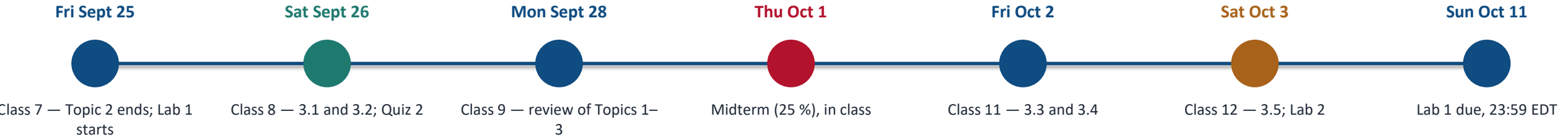
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Sections that apply today

- 6.1** authorization governs everything
- 6.4** lab rules: isolate, snapshot, fictional data, clean up
- 6.5** dual-use tools: lawful in your lab, an offence elsewhere
- 6.6** software licences and official sources
- 7.2** monitoring, capture and surveillance data
- 14 / C** external links: check before you click



Where we are



- From Topic 2 you have, for Ébénisterie Val-des-Pins Inc.: a network map, a register of **monitoring needs**, and an ordered **monitoring plan** (2,275 MB (megabyte) per day, 60 minutes of technician time per day).
- Topic 3 answers the next question: **with what?** Lesson 3.1 names the categories of tools; Lesson 3.2 chooses one tool for one need.
- The Midterm covers Topics 1 to 3 "up to the selection of tools" — that is exactly today's two lessons.
- Lab 2 (October 3) installs and configures the tool that Lesson 3.2 teaches you to choose.



What you will be able to do after this class

- 1 **Name the seven categories of monitoring tools**, say what kind of evidence each one observes, how it collects that evidence, and where it is placed on a small network.
- 2 **Classify any tool on six axes**: agent-based or agentless, passive or active, continuous or scheduled, signature-based or baseline-based, open-source or commercial, on-premises or cloud.
- 3 **Select a tool for a stated need with a repeatable procedure** — need, constraints, long list, knock-out filter, weighted scoring, pilot, decision record — and justify the choice in writing.

Competency EZ01, element 3, performance criterion 3.1 — "appropriate selection of surveillance tools". Today's coverage map and decision record, with the Lab 1 plan, are the three inputs of Lab 2.



The rule that governs every tool named today

- About twenty products are named. Each is lawful to study, to download from its **publisher's official site**, and to run inside your **Authorized Lab Environment** (the VirtualBox laboratory of the Safe Lab Walkthrough).
- Each is an offence to run against a network you do not own or are not authorized **in writing** to monitor.
- Section 7.2 of the Legal Notice, for this course:
 - capture only on **Designated Targets** inside the lab;
 - treat every capture, log file, baseline and alert record as **personal information**;
 - never install an agent, sensor or collector on a device belonging to another person, an employer or the College;
 - never enable promiscuous or monitor mode on a network you share with other people.
- **Nothing is installed today.** Today we recognize and choose; installation begins in Class 11, in the lab, on Designated Targets only.
- Links: official sites only, each with a [VirusTotal check](#) — Annex C: read the address letter by letter, scan it, then click.

Where a tool may run

Allowed your VirtualBox lab; Designated Targets; systems you own; systems with the owner's written authorization (Annex D)

Not allowed College networks; home or shared Wi-Fi; an employer's network; any third-party site or service

Rule if you are not certain you have authorization, you do not have it — stop and ask



Class 8 plan — 60 minutes of lecture

Time	Segment	Form	Planned
0:00 – 0:04	Where we are; today's objectives; the rule that governs every tool	Reading	4 min
0:04 – 0:20	Lesson 3.1 — what a monitoring tool is; the five kinds of evidence; the seven categories; six classification axes	Reading + Figures 3.1.1 and 3.1.2 + Table 3.1.A	16 min
0:20 – 0:28	Lesson 3.1 — real-life example (coverage map of Val-des-Pins) and its program, run step by step	Simulator + Table 3.1.B	8 min
0:28 – 0:34	Lesson 3.1 — flash cards and six questions	Interactive	6 min
0:34 – 0:46	Lesson 3.2 — the need first; the constraints; the seven-step procedure; frequent errors; four mini-cases	Reading + Figure 3.2.1	12 min
0:46 – 0:54	Lesson 3.2 — real-life example (weighted decision matrix for the Lab 2 tool) and its program	Simulator + Tables 3.2.A–C + Figure 3.2.2	8 min
0:54 – 0:58	Lesson 3.2 — flash cards and six questions	Interactive	4 min
0:58 – 1:00	Recap; Lab 2 preview; Quiz 2; Midterm reminder	Reading	2 min



Lesson 3.1 — Categories of monitoring tools

What each category observes, how it collects that evidence, and where it sits on a small network

30 minutes



3.1.1 What a monitoring tool is (1/2) — three functions

- A **monitoring tool** is a program, a device or a service that collects observations about a computer, a network device or a service, stores those observations, and presents or evaluates them.
- **Collection** — obtaining raw observations from a **data source**: an event log, a network interface, a counter inside a switch.
- **Storage** — keeping the observations for the **retention period** so that they can be searched later (Val-des-Pins: 90 days).
- **Evaluation** — comparing observations with a rule, a threshold or a **baseline** of normal activity, and producing an **alert** when the comparison indicates a problem.
- **Presentation** — showing observations to a person: table, graph, **dashboard**.
- A syslog server that only collects and stores is still a monitoring tool; a dashboard alone is useless without the other two.
- First question about any product: **which functions does it perform, and on which data source?**

The three functions at Val-des-Pins

Collect syslog from FW-01; Windows events from SRV-FILE-01

Store 90 days, within 200 GB, on MON-01

Evaluate alert when five logons fail in ten minutes

Present a dashboard the technician reads in 60 minutes a day

Ask which functions? on which data source?



3.1.1 What a monitoring tool is (2/2) — four components



- **Agent-based** tool: something is installed on the monitored machine (it sees inside).
- **Agentless** tool: the machine is observed from outside — syslog, SNMP (Simple Network Management Protocol), SPAN (Switched Port Analyzer) port, API (Application Programming Interface).
- The four words appear in every product's documentation; ask which components a product contains before comparing it with another.



3.1.2 Five kinds of evidence

Event

A record that something happened at a moment: a logon, a service stop, a firewall rule change. Written into logs by operating systems, applications and network devices.

Metric

A measured value sampled at regular intervals: processor load, free disk, bytes per second on an interface, whether a device answered a ping.

Flow record

The summary of one network conversation — addresses, protocol, ports, start time, packets and bytes — without the content.

Packet

The complete unit of data transmitted on the network, including its content when it is not encrypted.

State

The configuration, installed versions or list of known weaknesses of a system at one moment, obtained by inspecting it.

- Five kinds of evidence give the first five categories; two more categories examine the evidence for attack patterns (intrusion detection) or combine all of it (correlation and alerting).
- The monitoring needs of Class 6 were written with the data source in mind: each need points to a kind of evidence, and each kind of evidence points to a category.

Why categories exist

- Each tool observes **one kind of evidence**; the kinds differ in nature, in volume and in what they can prove.
- Tools of the **same category are largely interchangeable** — choosing between them is Lesson 3.2.
- Tools of **different categories are complementary** — one does not replace another, because they observe different evidence.
- Class 6 wrote needs with the data source in mind: "logon events on SRV-FILE-01" → events → category 1; "connection records at FW-01" → events written by the firewall → category 1; "device reachable" → metric → category 3.
- Two more categories collect no new evidence: **intrusion detection** examines traffic or host activity for signs of attack; the **SIEM** (Security Information and Event Management) combines everything.

Need → evidence → category

Logon events on SRV-FILE-01 event → C1

Device reachable (FW-01) metric → C3

Who talks to the Internet flow record → C2

Program never seen on PC-ACCT-01 host activity → C5

Known weakness on the server state → C6



Category 1 — Log collection and management (1/2)

C1

- Evidence: **events**. Function: collect and store; detects nothing by itself — it creates the searchable memory every other function depends on.
- **Windows Event Log** — channels Security, System, Application; the Security channel records logons, logoffs, account changes and, only when the **audit policy** enables it, access to files and folders.
- **syslog** — the format of RFC (Request for Comments) 5424 used by Linux and by network devices (firewalls, switches, access points, NAS (Network-Attached Storage)); sent to a **syslog server** on UDP (User Datagram Protocol) port 514 or, encrypted with TLS (Transport Layer Security), on TCP (Transmission Control Protocol) port 6514.
- Windows can forward events centrally with **WEF** (Windows Event Forwarding); more often an **agent** reads the local log and ships it.
- The platform **parses** (normalizes) each line into common fields, **indexes** it for fast search, and keeps it for the retention period.

C1 at a glance

Evidence events

Collects by agents on hosts; syslog from devices; WEF; cloud export

Output a searchable store with a retention period

Examples rsyslog, syslog-ng, Graylog, OpenSearch, Elastic Stack

Val-des-Pins all eight included assets



Category 1 — Log collection and management (2/2)

C1

- Examples: rsyslog and syslog-ng (collectors); [Graylog](#), [OpenSearch](#) with Fluent Bit, [Elastic Stack](#) (platforms).
- Val-des-Pins: FW-01, SW-01, SW-02 and NAS-01 send syslog; SRV-FILE-01 and PC-ACCT-01 send Windows events through an agent; MAIL-CLOUD exports its **audit log** through the provider's interface.
- Limit 1 — a log contains only what the **audit policy** was configured to record.
- Limit 2 — a compromised computer can alter or delete its local logs: copy them to a central store as soon as they are written.
- Limit 3 — every device needs the correct time from **NTP** (Network Time Protocol); otherwise "logon, then file access, then connection" cannot be reconstructed.
- Reference: NIST (National Institute of Standards and Technology) SP (Special Publication) 800-92, [Guide to Computer Security Log Management](#).

Val-des-Pins log sources

syslog FW-01, SW-01, SW-02, NAS-01

Agent SRV-FILE-01, PC-ACCT-01 (Windows events)

Export MAIL-CLOUD audit log

Records only CNC-PC-01 through FW-01

Three limits audit policy · central copy · NTP



Category 2 — Network traffic capture and analysis (1/2)

C2

- Evidence: **packets** and **flow records** — the network itself.
- A **packet analyzer** (protocol analyzer) such as [Wireshark](#) or [tcpdump](#) reads every frame on an interface and decodes its layers: Ethernet, IP (Internet Protocol), TCP or UDP, application.
- A computer receives only the frames addressed to it; to see other machines' traffic the switch must copy it:
 - **SPAN** (Switched Port Analyzer) or **mirror port** — a switch port that receives a copy of selected ports or VLANs (Virtual Local Area Networks);
 - **TAP** (Test Access Point) — a hardware device inserted in a cable that passes a copy to the sensor.
- **Full packet capture** stores every packet: the most complete evidence and the most expensive in disk space.
- Arithmetic: a 100 Mb/s office link busy 20 % of the time = 2.5 MB (megabyte) per second ≈ **216 GB (gigabyte) per day** — more than the whole 200 GB envelope of Val-des-Pins. That is why Class 7 deferred it.

C2 at a glance

Evidence packets and flow records

Collects by sensor on a SPAN port or TAP; flow export from FW-01

Output conversations: who, whom, when, how much; decoded packets

Examples Wireshark, tcpdump, ntopng, nfdump

Arithmetic $100 \text{ Mb/s} \times 20 \% = 2.5 \text{ MB/s} \approx 216 \text{ GB per day}$



Category 2 — Network traffic capture and analysis (2/2)

C2

- **Flow records** are the economical alternative: one record per conversation, exported by routers, firewalls and some switches in **NetFlow** (Cisco), **IPFIX** (IP Flow Information Export, RFC 7011) or **sFlow** (sampled) format.
- A **flow collector** — [ntopng](#), nfdump — stores and displays them; a firewall's own connection log is the same evidence written as events.
- Flow data answers "who talked to whom, when, on which port, how much" for months, in a small fraction of the space, and works even when the content is encrypted.
- Val-des-Pins: a sensor on a SPAN port of SW-01 observes the office; FW-01's connection records observe the Internet boundary; **CNC-PC-01** (no software allowed) is observed only this way.
- Limit — content protected by TLS is not readable.
- Legal boundary — a capture on a shared network records other people's communications: section 7.2 of the [Legal Notice](#) allows capture only in the lab, on Designated Targets; section 184 of the Criminal Code applies elsewhere.

One flow record

Source 192.0.2.10 (PC-ACCT-01)

Destination 203.0.113.5 : 443

Protocol TCP

Start 09:14:02

Packets / bytes 418 / 512,340

Content not kept — that is the point



Category 3 — Availability and performance monitoring (1/2)

C3

- Evidence: **metrics**. Two questions: is the device reachable, and how loaded is it?
- Method: **polling** — the server asks each device at a fixed interval (every 60 seconds, for example).
- **ICMP** (Internet Control Message Protocol) echo request — the **ping** — answers "reachable or not" and gives the round-trip time.
- **SNMP** (Simple Network Management Protocol) reads counters from a device's **MIB** (Management Information Base), each value identified by an **OID** (Object Identifier): bytes and errors per port, processor load, interface state.
- **SNMP version 3** adds authentication and encryption — the version to use; versions 1 and 2c send a **community string** (a password) in clear text.
- An **SNMP trap** is sent by the device itself when something happens (a link goes down), without waiting to be polled.
- Operating-system metrics (disk, memory, services) come from a light **agent** — the Zabbix agent, for example.

C3 at a glance

Evidence metrics: reachability, load, errors, disk

Collects by polling: ping; SNMP v3 on UDP port 161; traps on UDP port 162; light agents

Interval every 60 seconds at Val-des-Pins

Output green/red status map, graphs, threshold alerts

Examples Zabbix, Nagios Core, LibreNMS, Prometheus + Grafana



Category 3 — Availability and performance monitoring (2/2)

C3

- Examples: [Zabbix](#), [Nagios Core](#), [LibreNMS](#), [Prometheus](#) with [Grafana](#) dashboards; PRTG (commercial).
- Outputs: status maps (green reachable, red down), graphs over time, **threshold alerts** — "disk above 90 %", "no answer for 3 minutes".
- Val-des-Pins: FW-01, SW-01, SW-02, SRV-FILE-01 and NAS-01 are polled; the needs "service availability" and "disk usage" of Class 6 live here.
- Limit 1 — a metric says that something is wrong, rarely what.
- Limit 2 — polling is itself traffic and must be sized with the constraints of Class 6.
- Limit 3 — a device that is green (reachable) can be fully compromised.
- Availability tools are the first thing a small company installs, and the last thing that detects an intrusion.

Val-des-Pins polling

Polled FW-01, SW-01, SW-02, SRV-FILE-01, NAS-01

Thresholds disk above 90 %; no answer for 3 minutes

Needs served service availability; disk usage; port up/down

Remember green means reachable, not safe



Category 4 — Intrusion detection and prevention (1/2)

C4

- Evidence: traffic or host activity **compared with rules** that describe attacks.
- An **IDS** (Intrusion Detection System) detects and alerts; an **IPS** (Intrusion Prevention System) is placed **in-line** — the traffic passes through it — and can also block.
- A **NIDS** (Network-based Intrusion Detection System) such as [Suricata](#) or [Snort](#) reads packets from a SPAN port or a TAP and compares them with **signatures** — rules for known malicious patterns; rule sets are updated frequently (Lesson 3.3).
- [Zeek](#) is an **NSM** (Network Security Monitoring) tool: it turns packets into structured logs (connections, DNS (Domain Name System) requests, TLS certificates) — a bridge between category 2 and category 1.
- A **HIDS** (Host-based Intrusion Detection System) such as [Wazuh](#) or OSSEC runs as an agent and examines the host's logs, file integrity and signs of hidden software.

C4 at a glance

Evidence traffic or host activity compared with rules

Placement NIDS on the SPAN sensor (passive) or IPS in-line; HIDS agent on hosts

Output alerts with a rule name and a severity

Examples Suricata, Snort, Zeek, Wazuh/OSSEC

Reference NIST SP 800-94



Category 4 — Intrusion detection and prevention (2/2)

C4

- **Signature-based detection** recognizes known patterns: precise, and blind to an attack that has never been described.
- **Anomaly-based (baseline-based) detection** flags deviation from measured normal activity: catches new behaviour, produces more false alarms. Both return in Topic 5.
- Reference: NIST SP 800-94, [Guide to Intrusion Detection and Prevention Systems](#).
- Val-des-Pins: the NIDS runs on the same sensor as category 2, fed by the SPAN port of SW-01; the HIDS agent runs on SRV-FILE-01 and PC-ACCT-01.
- Limits: encrypted traffic hides content from a NIDS; false positives consume the technician's 60 minutes; rules must be updated.
- An IPS that blocks legitimate traffic is a **defensive change** under section 7.4 of the Legal Notice — approved, with a window and a rollback, or not at all.

Two detection methods

Signature-based known patterns; precise; blind to new attacks

Anomaly-based deviation from the baseline; catches novelty; more false alarms

Val-des-Pins NIDS on SW-01's SPAN sensor; HIDS on SRV-FILE-01 and PC-ACCT-01

Caution an in-line IPS is a defensive change — section 7.4



Category 5 — Endpoint monitoring

C5

- Evidence: **activity inside the computer** — process creation with command line and parent, file creation or modification, registry changes, network connections per program, logons, removable media.
- Windows: [Sysmon](#) (System Monitor, free Microsoft Sysinternals component) writes detailed events into the Event Log; Linux: auditd; an agent ships them centrally.
- **EDR** (Endpoint Detection and Response) adds response actions, such as isolating the computer; **FIM** (File Integrity Monitoring) detects any change to a list of critical files; antivirus and antimalware alerts are endpoint evidence too.
- Val-des-Pins: PC-ACCT-01 (highest exposure to people and media in Class 7) and SRV-FILE-01; **never CNC-PC-01** — the vendor's support contract forbids installing software on it.
- Limits: one installation and one update cycle per computer; processor and memory use; detailed data about how a person uses a computer — data minimization under Law 25 and section 6.7 is a design decision, not an afterthought.

C5 at a glance

Evidence processes, files, registry, logons, removable media

Collects by agent: Sysmon + shipper, Wazuh agent, EDR

Output detailed host events; response actions

Val-des-Pins PC-ACCT-01 and SRV-FILE-01; never CNC-PC-01

Privacy collect the minimum — Law 25, section 6.7



Category 6 — Configuration and vulnerability assessment

C6

- Evidence: **state at one moment**, not activity.
- A **vulnerability scanner** — [Greenbone \(OpenVAS\)](#), or Nessus as a commercial example — connects to each system, identifies the services and versions that answer, and compares them with a database of known weaknesses identified by **CVE** (Common Vulnerabilities and Exposures) numbers.
- A **configuration assessment** tool — CIS-CAT or any **SCAP** (Security Content Automation Protocol)-compatible tool — compares settings with a published **benchmark** from the [CIS](#) (Center for Internet Security).
- Both run on a schedule (weekly, monthly) and both are **active**: they send packets, create load, and appear in the target's logs exactly as the first phase of an attack would.
- In this course you only recognize the category; scanning is taught in 420-3C1-DW Managing Vulnerabilities — section 7.6 applies: even inside the lab, only Designated Targets.
- Val-des-Pins: a monthly state check of SRV-FILE-01 and FW-01 is a later addition, not part of Lab 2.

C6 at a glance

Evidence state: versions, settings, known weaknesses (CVE)

Collects by scheduled active scan; benchmark comparison (SCAP)

Output findings with a severity; compliance score

Examples Greenbone (OpenVAS), CIS-CAT; Nessus

This course recognize only — section 7.6; scanning is 420-3C1-DW



Category 7 — Correlation, alerting and visualization (SIEM)

C7

- A **SIEM** (Security Information and Event Management) receives events, flow records, metrics and alerts from the other categories, normalizes them, stores them and **correlates** them.
- A **correlation rule** connects observations that are harmless one by one: five failed logons, then one successful logon, from the same address, within ten minutes.
- Alerts carry a **severity**; dashboards, reports and **case management** (who examined which alert, what was concluded — Lesson 5.4).
- Examples: [Wazuh](#) (agents + manager + dashboard, open-source); [Security Onion](#) (Suricata, Zeek, host agents and a search platform in one distribution); [Elastic Security](#) and Graylog; Splunk and cloud SIEM services (commercial).
- A SIEM is only as useful as its sources; for a 22-person company it is justified from **three or more sources** — Val-des-Pins reaches that at Lab 2 (firewall, file server, accounting workstation).
- Limits: commercial pricing by data volume; skills; and **tuning** — the reduction of false alarms that occupies Topic 6.

C7 at a glance

Evidence everything from C1 to C6

Function normalize, store, correlate, alert, dashboards, cases

Rule example 5 failed logons then 1 success, same address, 10 minutes

Examples Wazuh, Security Onion, Elastic Security, Graylog; Splunk, cloud SIEM

Val-des-Pins the MON-01 console — justified from three sources (Lab 2)



MON-01 is one lab virtual machine hosting several categories; arrows show the direction of the data (polling goes from the server to the device). Schematic, not a vendor screenshot.



3.1.4 Six ways to classify any tool

- **Agent-based** — software on the monitored machine, sees inside | **Agentless** — observes from outside: syslog, SNMP, SPAN port, API (Application Programming Interface); the only option for CNC-PC-01.
- **Passive** — receives logs or a copy of the traffic, adds no packets | **Active** — sends pings, SNMP polls, scans; creates load and is recorded in the target's logs.
- **Continuous** — observes all the time, delay of seconds to minutes | **Scheduled** — runs at set times, shows the state at that moment.
- **Signature-based** — matches known patterns | **Baseline-based** — flags deviation from measured normal activity.
- **Open-source** — code published, no licence cost, community or paid support | **Commercial** — licence per device or per data volume, vendor support included.
- **On-premises** — runs on the organization's own hardware or lab VM (Virtual Machine), data stays inside | **Cloud / SaaS** (Software as a Service) — runs on the provider's servers, data leaves the site, subscription cost.
- Each axis has consequences for cost, network load, privacy and skills.

Val-des-Pins on the six axes

Agent PC-ACCT-01 yes; CNC-PC-01 no (contract)

Passive / active sensor passive; poller and scanner active

Timing logs continuous; scans scheduled

Detection IDS signatures; Topic 3.4 baseline

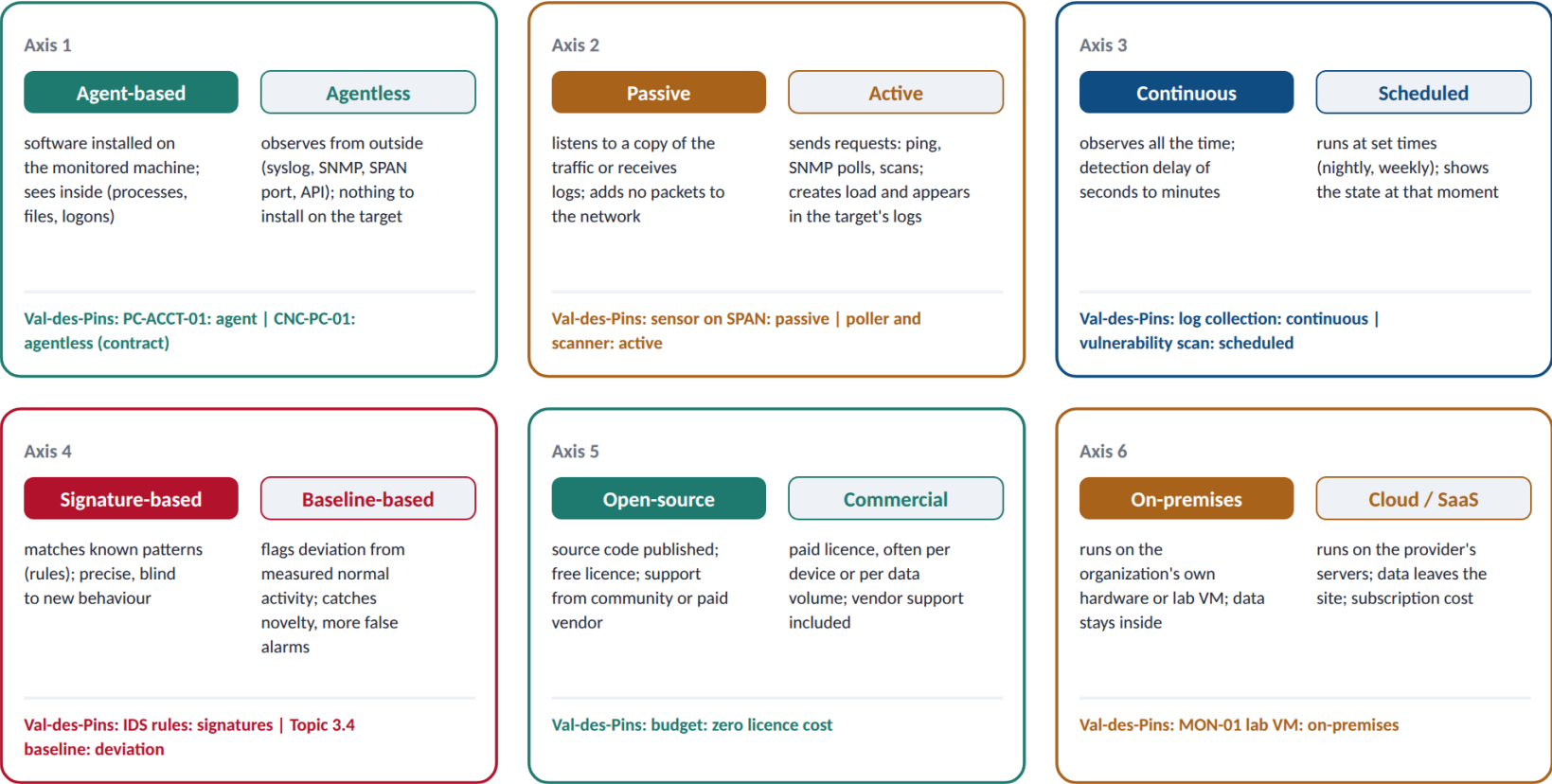
Licence zero cost — open-source

Location MON-01 on-premises, in the lab



Figure 3.1.2 — Six ways to classify a monitoring tool

Figure 3.1.2 - Six ways to classify a monitoring tool (every tool sits on one side of each axis)



Cyber.SoHo Educational Hub - 420-1C1-DW - Class 8 - Figure 3.1.2

Every tool sits on one side of each axis; the Val-des-Pins position is shown under each pair.



3.1.5 How the categories combine — the coverage map

- A monitoring need is **covered** when at least one category observes the evidence it requires; a plan is covered when every included need is.
- The **coverage map**: assets in rows, categories in columns, a mark in each cell where a need is served.
- A critical asset should be observed by **two independent categories** — the file server by its own event log (C1) and by the flow records of the switch (C2) — so that the failure or compromise of one source does not blind the technician.
- One machine can host several categories: **MON-01**, a single lab virtual machine, runs the log collector, the availability poller, the network sensor and the SIEM console at the same time.
- Categories are **functions, not boxes**.

Coverage rules

Covered need at least one category observes its evidence

Covered plan every included need is covered

Critical asset two independent categories

One machine MON-01 hosts C1, C3, C2/C4, C5 manager and C7

Empty cell a candidate improvement for Lesson 6.3 (Lab 3)



Table 3.1.A — Reference table of the seven categories (1/2)

Code	Category	Evidence observed	Collection method	Typical outputs	Examples (open-source; commercial)	Placement at Val-des-Pins
C1	Log collection and management	Events (Windows Event Log, syslog, cloud audit logs)	Agent on hosts; syslog from devices; API export from cloud; WEF	Searchable store of events, retention	rsyslog, syslog-ng, Graylog, OpenSearch + Fluent Bit, Elastic Stack	All included assets; MAIL-CLOUD via audit export
C2	Network traffic capture and analysis	Packets (full capture) and flow records (NetFlow, IPFIX, sFlow)	Sensor on a SPAN port or TAP; flow export from FW-01	Conversations: who, whom, when, how much; decoded packets	Wireshark, tcpdump, ntopng, nfdump	SPAN port of SW-01; FW-01 connection records; CNC-PC-01 (agentless)
C3	Availability and performance monitoring	Metrics (reachability, load, errors, disk)	Polling with ICMP and SNMP v3; traps; light agents	Status maps, graphs, threshold alerts	Zabbix, Nagios Core, LibreNMS, Prometheus + Grafana; PRTG (commercial)	FW-01, SW-01, SW-02, SRV-FILE-01, NAS-01
C4	Intrusion detection and prevention	Traffic or host activity compared with rules	NIDS on the sensor (passive) or in-line IPS; HIDS agent	Alerts with a rule name and severity	Suricata, Snort, Zeek (NSM), Wazuh/OSSEC (HIDS)	Sensor on SW-01; agents on SRV-FILE-01 and PC-ACCT-01

Product names are used for identification only; Cyber.SoHo is not affiliated with any vendor (Legal Notice, section 5). Verify each product's current licence and requirements on its official site.



Table 3.1.A — Reference table of the seven categories (2/2)

Code	Category	Evidence observed	Collection method	Typical outputs	Examples (open-source; commercial)	Placement at Val-des-Pins
C5	Endpoint monitoring	Host activity: processes, files, registry, logons, removable media	Agent (Sysmon + shipper, Wazuh agent, EDR)	Detailed host events; response actions (EDR)	Sysmon, auditd, osquery, Wazuh agent; commercial EDR	PC-ACCT-01, SRV-FILE-01; never CNC-PC-01 (contract)
C6	Configuration and vulnerability assessment	State: versions, settings, known weaknesses (CVE)	Scheduled active scan; benchmark comparison (SCAP)	Findings list with severity; compliance score	Greenbone (OpenVAS), CIS-CAT; Nessus (commercial)	Later, lab only, Designated Targets (s. 7.6) — not in Lab 2
C7	Correlation, alerting and visualization (SIEM)	All of the above	Receives from C1–C6; normalizes, correlates, alerts	Dashboards, correlated alerts, cases, reports	Wazuh, Security Onion, Elastic Security, Graylog; Splunk and cloud SIEM (commercial)	MON-01 console; justified from three sources (Lab 2)

Videos for Lesson 3.1 (after class; each under 15 minutes)

Logs and Monitoring – CompTIA Network+ N10-009 – 3.2

Professor Messer (Messer Studios) · length 13:06 (under 15 minutes)

Categories 1, 2 and 7 in thirteen minutes: flow data and its collector, the protocol analyzer, syslog with its facility and severity codes, the SIEM console, and the port mirror (SPAN) that feeds a sensor — the placement of Figure 3.1.1.

[Open on YouTube](#) [VirusTotal check of the link](#) <https://www.youtube.com/watch?v=ieqSi5Aicxc>

Security Tools – CompTIA Security+ SY0-701 – 4.4

Professor Messer (Messer Studios) · length 14:06 (under 15 minutes)

The security side of the same categories: SCAP and benchmarks (C6), agents versus agentless (axis 1), SIEM (C7), antivirus and data loss prevention (C5), SNMP traps (C3), NetFlow (C2) and vulnerability scanners (C6).

[Open on YouTube](#) [VirusTotal check of the link](#) <https://www.youtube.com/watch?v=nNINTviiacU>

Links verified on September 23, 2026 against the publisher's course index. Third-party content — Legal Notice, sections 14 and 15.5.



Real-life example 3.1 — coverage map of Val-des-Pins (1/2): the task

- **Task.** Take every asset kept in the Class 7 plan, take each of its monitoring needs, identify the kind of evidence the need requires, and map that evidence to one of the seven categories.
- **Produce:** a coverage map (assets in rows, categories in columns) and a spreadsheet listing every need with its category and its collection method (agent or agentless).
- **How it relates to the lesson.** Lesson 3.1 applied to a real plan: it turns the "what to observe" of Topic 2 into the "which category of tool" of Topic 3, and shows why a small network needs several categories on one machine.
- **Where it goes next.** Section 1 of the **Tool Selection Record** handed in with Lab 2 (October 3); the categories serving the Tier 1 assets decide the first tool installed in Lab 2; reused in Lab 3 (October 16) for Lesson 6.3 — every empty cell is a candidate improvement.

Situation

When Monday, September 28, 2026 — one hour before the shop opens

Where Ébénisterie Val-des-Pins Inc., Laval (fictional)

Who the part-time technician

Inputs Lab 1 map; Class 6 register of needs; Class 7 plan

Output coverage map + CSV + gap list

Feeds Lab 2 (Oct 3) and Lab 3 (Oct 16)



Real-life example 3.1 — coverage map of Val-des-Pins (2/2): the situation

- Monday, September 28, 2026, one hour before the shop opens. On the technician's desk: the Lab 1 network map, the Class 6 register of needs, the Class 7 plan — FW-01, MAIL-CLOUD, PC-ACCT-01, SW-01, SRV-FILE-01, CNC-PC-01 (connection records only), SW-02, NAS-01 (override); AP-01 and the office PCs deferred.
- The owner asks: "Which software do we need to buy?" The answer must begin with **categories, not brand names** — twenty-three needs of five kinds of evidence.
- Need by need: "administrator logins on the firewall" → syslog → C1; "device reachable" → metric by polling → C3; "program never seen before on PC-ACCT-01" → host activity → C5 (agent); "copy of office traffic" → packets from the SPAN port of SW-01 → C2, feeding C4.
- CNC-PC-01: "logon events" needs a Windows event log, which needs an agent, which the contract forbids → recorded as a **gap**; the asset stays observed agentlessly through FW-01's connection records.
- Deliverables:** vdp_tool_category_map.csv (one row per need), the coverage matrix (Figure 3.1.3) with one paragraph — Tier 1 assets need **C1, C3 and C5** — and the list of gaps with the constraint that causes each.

Result

Assets 8 included, 23 needs

Categories used C1, C3, C5, C2, C4

Tier 1 needs C1 + C3 + C5

Not yet C6 (later, lab only); C7 console on MON-01

Gap CNC-PC-01 logon events — no agent by contract



Program 3.1 — class8_tool_categories.py (Python 3, no external modules)

1

Constants. CATEGORIES (C1–C7) and EVIDENCE_TO_CATEGORY — Lesson 3.1.2 written as a dictionary; AGENT_REQUIRED names the two kinds of evidence that need software on the machine.

2

The plan. PLAN is the Class 7 order: eight assets, each with its tier, its agent_allowed flag and its needs as (need, evidence) pairs — twenty-three needs.

3

Build the map. For each asset and each need: look up the category, decide whether an agent is needed; agent needed and not allowed → record a gap and continue; otherwise mark the cell and print the mapping.

4

Summarize. Count the assets served by each category, sort from the most used, print a bar of # characters.

5

Results for Lab 2. Categories required by the plan: C1, C3, C5, C2, C4; categories serving Tier 1 assets: C1, C3, C5; one gap (CNC-PC-01); write vdp_tool_category_map.csv.

6

In the lecture page: the simulator runs the same logic step by step — Start / Pause / Step / Restart, a trace window, and the matrix filling in; a comment explains every line; Copy and Download buttons.

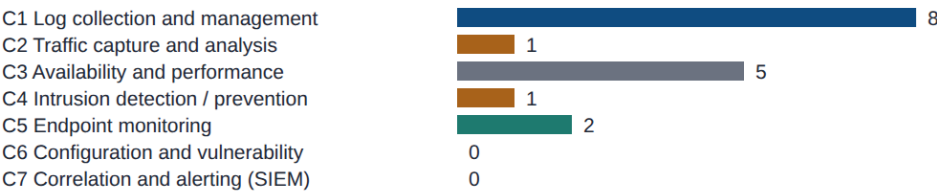


Figure 3.1.3 — The coverage matrix of Val-des-Pins

Figure 3.1.3 — Coverage matrix: assets × categories (fills in as the program runs)

	C1	C2	C3	C4	C5	C6	C7
FW-01 (T1)	✓		✓				
MAIL-CLOUD (T1)	✓						
PC-ACCT-01 (T1)	✓				✓		
SW-01 (T2)	✓	✓	✓	✓			
SRV-FILE-01 (T2)	✓		✓		✓		
CNC-PC-01 (T2)	✓						
SW-02 (T3)	✓		✓				
NAS-01 (T3)	✓		✓				

Assets served per category



Lab 2 first tool must cover C1 + C3 + C5 — gap: CNC-PC-01 stays agentless

Result: Tier 1 assets need C1 + C3 + C5 → the first Lab 2 tool must collect logs, poll availability and manage agents (or be a platform doing all three). Gap: CNC-PC-01 logon events (no agent by contract).

Cells: navy C1, ochre C2/C4, grey C3, teal C5; the red dot on CNC-PC-01 is the documented gap. Downloadable from the lecture page (SVG/PNG).



Table 3.1.B — Coverage map, one line per asset

Asset	Tier	Categories covering its needs	Collection	Gap
FW-01	1	C1, C3	agentless	—
MAIL-CLOUD	1	C1	agentless	—
PC-ACCT-01	1	C1, C5	agent	—
SW-01	2	C1, C3, C2, C4	agentless	—
SRV-FILE-01	2	C1, C3, C5	agent + agentless	—
CNC-PC-01	2	C1	agentless	GAP: logon events on the machine PC
SW-02	3	C1, C3	agentless	—
NAS-01	3	C1, C3	agentless	—

Fictional organization. Reuse in Lab 2 (tool selection) and Lab 3 (Lesson 6.3, improving coverage).



Flash cards — Lesson 3.1

1

The three functions of a monitoring tool

Collection (obtain observations from a data source), storage (keep them for the retention period), evaluation and presentation (compare with rules, thresholds or a baseline; show and alert).

2

Agent versus sensor

An agent is a program installed on the monitored computer that reads local data (logs, processes, files). A sensor observes network traffic from outside the computer, fed by a SPAN port or a TAP; nothing is installed on the target.

3

The five kinds of evidence

Events (logs), metrics (sampled values), flow records (conversation summaries), packets (complete transmitted units) and states (configuration or weaknesses at one moment). Two further categories examine or combine them: intrusion detection and SIEM.

4

SPAN port and TAP

A SPAN (mirror) port is a switch port configured to receive a copy of the frames of selected ports or VLANs so that a sensor can see them. A TAP is a hardware device inserted into a cable that passes a copy to the sensor without depending on the switch.

5

SIEM

Security Information and Event Management: a platform that receives events, flow records, metrics and alerts from the other categories, normalizes and stores them, correlates them with rules (five failed logons then one success within ten minutes) and raises alerts with a severity. Justified from about three sources.

Questions — Lesson 3.1 (1/2)

1

Which category of monitoring tool observes flow records — who talked to whom, when, on which port and how much — without storing the content of the packets? (MCQ)

- a) Availability and performance monitoring (C3)
- b) Network traffic capture and analysis (C2), with a flow collector
- c) Endpoint monitoring (C5)
- d) Configuration and vulnerability assessment (C6)

2

The technician wants to know within one minute whether SRV-FILE-01 has stopped answering. Which category answers this need most directly? (MCQ)

- a) Log collection and management (C1)
- b) Intrusion detection (C4)
- c) Availability and performance monitoring (C3), polling with ICMP or SNMP every minute
- d) Correlation and alerting (C7) alone

3

What is the difference between an agent and a sensor? (MCQ)

- a) An agent is installed on the monitored computer and reads its local data; a sensor observes network traffic from outside the computer, fed by a SPAN port or a TAP
- b) An agent is hardware; a sensor is software
- c) An agent collects metrics only; a sensor collects logs only
- d) There is no difference; the words are synonyms

Answers and the explanation of every option are in the speaker notes and in the lecture page (Check / Reveal buttons).



Questions — Lesson 3.1 (2/2)

4 Why does the Val-des-Pins coverage map record a gap for the need 'logon events on CNC-PC-01'? (MCQ)

- a) CNC-PC-01 runs an operating system that has no event log
- b) The vendor's support contract forbids installing software on the machine, so the agent needed to read the Windows event log cannot be installed; the asset is observed agentlessly through FW-01's connection records
- c) Endpoint agents do not exist for industrial computers
- d) The machine is not connected to the network

5 Which of the following are limits of log collection and management tools (category 1)? (select 3) (MSQ – select 3)

- a) A log contains only what the audit policy was configured to record
- b) Logs cannot be searched once they are stored
- c) Without synchronized clocks (NTP), events from different devices cannot be placed in order
- d) A compromised computer can alter or delete its local logs, which is why logs are copied to a central store
- e) Log collection requires a SPAN port on every switch
- f) Log management only exists as commercial software

6 Which statements about intrusion detection and prevention systems (category 4) are correct? (select 3) (MSQ – select 3)

- a) A NIDS reads traffic from a SPAN port or a TAP
- b) An IPS is placed in-line and can block traffic
- c) Signature-based detection catches previously unknown attacks better than anomaly-based detection
- d) A HIDS runs as an agent on the host and examines its logs and file integrity
- e) An IDS can read the content of traffic protected by TLS
- f) Rule sets never need updating once installed

Answers and the explanation of every option are in the speaker notes and in the lecture page (Check / Reveal buttons).



Lesson 3.2 — Selecting the right tool for a given need

The need first, the constraints second, the tool last — a repeatable seven-step procedure with a decision record

24 minutes



3.2.1 The need comes first (1/2) — six elements

- A tool is selected to satisfy a **stated need**; a need is never rewritten to fit a tool that someone already likes.
- A testable need has six elements: the **asset** observed; the **evidence** required (event, metric, flow record, packet, state); the **behaviour** to detect; the acceptable **delay** between behaviour and alert; the **recipient** of the alert; the **retention** of the data.
- **Need N-01 (Lab 2):** "collect the security event logs of FW-01, SRV-FILE-01, PC-ACCT-01 and SW-01, keep them 90 days within 200 GB, and alert the technician by e-mail within 5 minutes when one of the predefined suspicious behaviours occurs."
- Every word of that sentence can be verified during the pilot — a need that cannot be tested cannot be used to choose.

N-01 in six elements

Asset FW-01, SRV-FILE-01, PC-ACCT-01, SW-01

Evidence security event logs (events)

Behaviour one of the predefined suspicious behaviours

Delay alert within 5 minutes

Recipient the technician, by e-mail

Retention 90 days, within 200 GB



3.2.1 The need comes first (2/2) — requirements, preferences, knock-outs, weights

- A **requirement** is imposed by the need or by a constraint — "reads Windows event logs".
- A **preference** is something the technician would like — "a dark-themed dashboard". Requirements decide; preferences only break ties.
- A **knock-out criterion** (must-have) is a condition whose failure removes a candidate whatever its other qualities.
- A **weighted criterion** is a quality that candidates satisfy more or less and that is scored.
- Confusing the two produces the classic error: a spreadsheet in which an excellent but unaffordable product "wins".

Examples from N-01

Requirement reads Windows event logs and syslog

Preference a dark-themed dashboard

Knock-out zero licence cost; fits the 8 GB lab VM

Weighted ease of use for a beginner (weight 20)

Classic error the excellent but unaffordable product wins



3.2.2 The constraints that filter the list

- **Budget and licence** — no licence budget: open-source software or editions free for this use; "free" is verified on the publisher's site (limits on data volume or devices).
- **People and skills** — one part-time technician, 60 minutes per day; a platform needing two hours of daily care will stop being used.
- **Hardware capacity** — MON-01 is one lab VM with 8 GB of RAM (Random-Access Memory) and 200 GB of disk; the publisher's documented minimum decides.
- **Network impact** — agents use processor time, pollers create traffic, a SPAN port occupies a switch port.
- **Operating systems and sources** — Windows 11 event logs, Linux and device syslog, cloud audit export: a tool reading only one does not cover the need.
- **Contracts** — CNC-PC-01 must remain agentless.
- **Law and privacy** — data minimization and a retention limit (section 7.2, Law 25).
- **Documentation and support, maintenance** (updates, rule sets, storage growth — Lesson 3.3), **exit** (removable, data exportable, no lock-in).

Val-des-Pins in numbers

Licence budget 0 CAD

Technician time 60 minutes per day

MON-01 8 GB RAM, 200 GB disk

Envelope 2,275 MB per day (Class 7)

Sources Windows 11, Linux, devices, cloud export

Contract CNC-PC-01 agentless



3.2.3 The selection procedure (1/2) — steps 1 to 4

- 1 **State the need** with its six elements.
- 2 **List the constraints** that apply to this need — most come from Lesson 2.3 and from the envelope of Lesson 2.5.
- 3 **Build the long list** — candidates from the category the need requires (Lesson 3.1); a need for events is not served by a packet analyzer, however good. Four to eight candidates.
- 4 **Apply the knock-out filter** — verify each must-have from the publisher's documentation, not from marketing pages; one failure removes the candidate and the reason is written down. The survivors form the **short list**.



3.2.3 The selection procedure (2/2) — steps 5 to 7

5

Weighted scoring — list the criteria and fix the weights (sum 100) **before** looking at any candidate; score each short-listed candidate 0 to 5 per criterion with one sentence of justification. This is a **decision matrix**; its value lies in the written justifications.

6

Pilot — install the first-ranked candidate in the Authorized Lab Environment for a fixed time; measure what the need says: installation time, memory and disk per day, whether each predefined suspicious behaviour raises an alert within the delay; snapshot before, rollback ready. If it fails, pilot the next candidate.

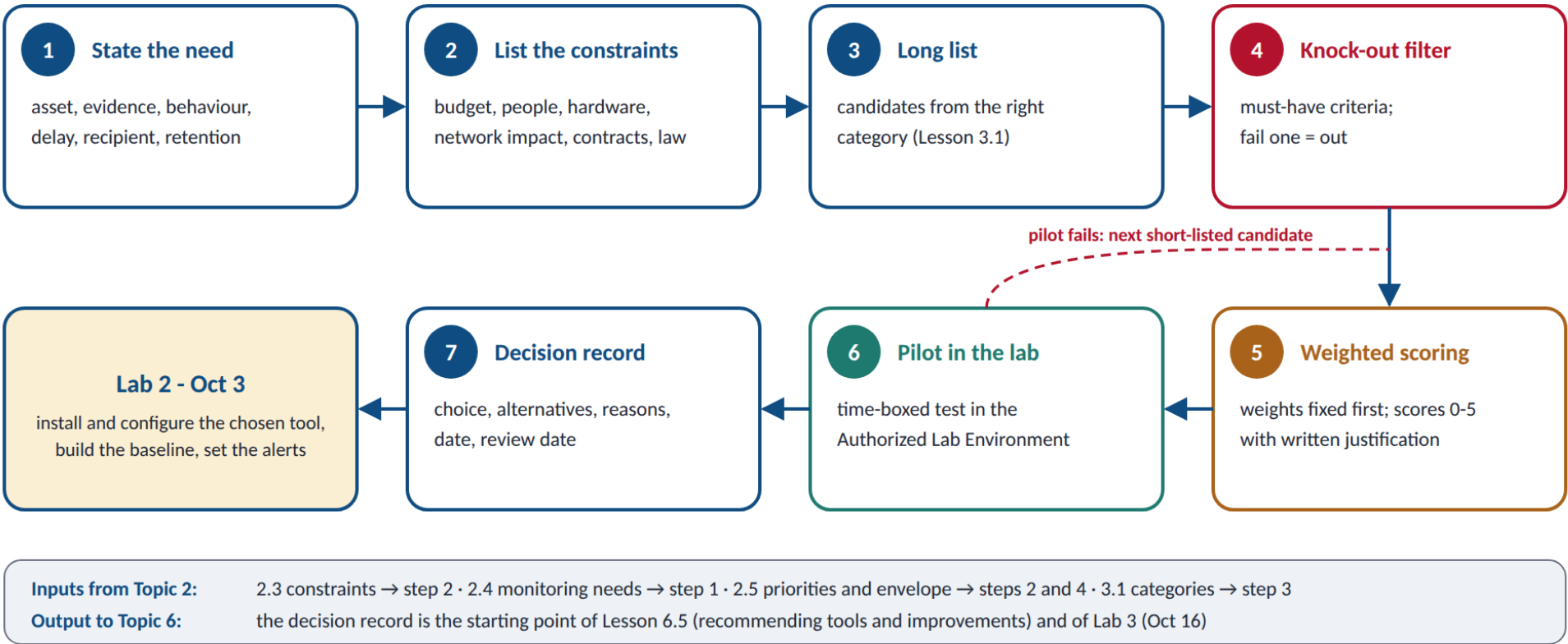
7

Decision record — one page: need, constraints, candidates, knock-outs, scores, choice, alternatives rejected and why, date, **review date**. What you show an auditor, a successor, or the owner who asks "why this one?" — and the start of Lesson 6.5.

Weighted total = $\Sigma (\text{weight} \times \text{score}) \div 5$ — with weights summing to 100 and scores from 0 to 5, the total is a percentage.

Figure 3.2.1 — The seven-step selection procedure

Figure 3.2.1 - The tool selection procedure: the need is stated first, the tool is chosen last



Cyber.SoHo Educational Hub - 420-1C1-DW - Class 8 - Figure 3.2.1

Inputs from Topic 2 (constraints, needs, priorities) and Lesson 3.1 (categories); output to Lab 2 and to Lesson 6.5 / Lab 3.



3.2.4 Frequent selection errors

- **Choosing by popularity** — the most downloaded tool is not the one that fits a 22-person company; popularity says something about documentation and community, nothing about fit.
- **Resource mismatch** — a documented minimum of 16 GB of RAM on an 8 GB lab machine: checked at the knock-out step, not discovered in the pilot.
- **Ignoring licence terms** — "free" editions with a daily data or device limit stop working, or become illegal to use (section 6.6).
- **Forgetting the technician's time** — a tool is chosen for the ninety days that follow, not for the day of installation.
- **Collecting more than the need** — full packet capture "because it is more complete" exceeds storage and records content nobody asked for.
- **No rollback** — a pilot without a snapshot turns a failed test into a broken lab.
- **Wrong category** — an availability monitor that "also has a log item" is not a log platform.
- **No decision record** — an undocumented choice cannot be reviewed, defended or improved.

Check yourself

Category? does the tool observe the evidence the need requires?

Fits? documented minimum against the lab VM

Licence? verified on the publisher's site

Time? ninety days of maintenance, not one day of installation

Rollback? snapshot before the pilot

Written? decision record with a review date



3.2.5 Four mini-cases — the need decides the category

- **A — "Within five minutes if the file server stops answering."** Metric → C3 availability monitoring: poll SRV-FILE-01 every 60 s with ICMP, read disk counters with SNMP v3 or a light agent (Zabbix, Nagios Core, LibreNMS). A packet analyzer shows the absence of traffic only to someone watching; an IDS has nothing to compare.
- **B — "Alert me on five failed logons to PC-ACCT-01 in ten minutes."** Windows Security events 4625 and 4624 → C1 log collection with a correlation rule (C7): Wazuh, Graylog with a shipper, Elastic Security. The audit policy must record failed logons, or the tool sees nothing.
- **C — "Which internal computer sends the most data to the Internet each day?"** Flow records or FW-01's connection records → C2 flow collector (ntopng) or a report on the firewall log (C1). Full packet capture would cost a hundred times the storage and record content nobody asked for.
- **D — "Observe CNC-PC-01 without installing anything on it."** Its connections → agentless: FW-01 connection records (C1) and, if needed, the SPAN port of SW-02 feeding the sensor (C2, C4). The vendor contract removes every agent-based candidate at the knock-out step.

The need decides the category

A — up or down? metric → C3 availability monitoring

B — five failed logons events → C1 + correlation rule (C7)

C — who sends the most flow records → C2 (or FW-01 log, C1)

D — nothing on CNC-PC-01 agentless → FW-01 records (C1); SPAN sensor (C2, C4)



Table 3.2.A — Knock-out and weighted criteria for need N-01

Code	Kind	Criterion	Weight	Origin of the criterion
K1	Knock-out	Belongs to category C1/C7 — log collection with rule-based alerting (the category the need requires)	—	Wrong category is the most frequent error; checked first
K2	Knock-out	Zero licence cost for this use	—	Constraint: no licence budget (Lesson 2.3)
K3	Knock-out	Documented minimum RAM fits the 8 GB lab VM	—	Constraint: hardware capacity of MON-01
K4	Knock-out	Can send an e-mail alert when a rule matches	—	Need N-01 requires an alert within 5 minutes
W1	Weighted	Fit to the need: log collection plus rule-based alerts	25	The largest weight goes to the need itself
W2	Weighted	Ease of installation and daily use for a beginner	20	One part-time technician; Lab 2 is done by students
W3	Weighted	Resource demand inside the lab VM (memory, disk)	15	Envelope: 8 GB RAM, 200 GB disk, 2,275 MB/day
W4	Weighted	Documentation and community support	10	The technician will search the documentation daily
W5	Weighted	Daily maintenance effort (updates, rules, storage)	10	60 minutes per day, all monitoring included
W6	Weighted	Coverage: Windows, Linux, network devices, cloud logs	10	Sources listed in the need and in Table 3.1.B
W7	Weighted	Growth: can add flow records or an IDS later	10	Lab 3 and Lesson 6.3 will extend coverage

Videos for Lesson 3.2 (after class; each under 15 minutes)

Network Solutions – CompTIA Network+ N10-009 – 3.2

Professor Messer (Messer Studios) · length 6:57 (under 15 minutes)

Matching a solution to a need: device discovery, traffic analysis from a firewall log (mini-case C), performance statistics, availability monitoring with green/red status and alerts (mini-case A), configuration monitoring.

[Open on YouTube](#) [VirusTotal check of the link](#) https://www.youtube.com/watch?v=zIX_SzezCi0

SNMP – CompTIA Network+ N10-009 – 3.2

Professor Messer (Messer Studios) · length 8:58 (under 15 minutes)

The agentless method behind category 3 and mini-case A: MIB, OID, polling on UDP port 161, traps on UDP port 162, community strings in versions 1 and 2c, authentication and encryption in version 3.

[Open on YouTube](#) [VirusTotal check of the link](#) <https://www.youtube.com/watch?v=We5MkaEJOs0>

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Real-life example 3.2 — choosing the Lab 2 tool (1/2): the task

- **Task.** For need N-01, apply the seven steps: state the need, list the constraints, long list of six candidates, four knock-out criteria, weighted scoring of the survivors on seven criteria, decision record with a pilot plan and a review date.
- **Produce:** the decision matrix as a spreadsheet and the decision record as a one-page table.
- **How it relates to the lesson.** Lesson 3.2.3 executed on the coverage map of Lesson 3.1: the Tier 1 categories (C1, C3, C5) define the long list; the constraints of 3.2.2 define the knock-outs; Table 3.2.A defines the score.
- **Where it goes next.** Sections 2 and 3 of the Tool Selection Record of Lab 2 (October 3); the chosen tool is the one you install and configure in Lab 2; the review date, October 16, is Lab 3, where Lesson 6.5 asks you to recommend tools and improvements from this record.

Situation

When Monday, September 28, 2026 — one hour later

Need N-01 (first tool for Lab 2)

Long list 6 candidates

Knock-outs K1 category · K2 cost · K3 RAM · K4 alerting

Weights 25 / 20 / 15 / 10 / 10 / 10 / 10 = 100

Output decision matrix (CSV) + decision record



Real-life example 3.2 — choosing the Lab 2 tool (2/2): the situation

- The coverage map says the first tool must collect Windows and device logs, manage endpoint agents and raise alerts; availability polling can be a second tool later.
- Long list of six: **Wazuh** and **Graylog Open** (open-source, logs and alerts); **Security Onion** (open-source network-security distribution); **Zabbix** (availability monitor); **Wireshark** (packet analyzer); a commercial **cloud SIEM** subscription.
- Four facts per candidate from each publisher's documentation: category, licence cost, documented minimum memory, e-mail alerting.
- Knock-outs remove Security Onion (K3 — memory above the 8 GB lab VM), Zabbix (K1 — category C3), Wireshark (K1, K4 — no central store, no alerting) and the cloud SIEM (K2 — monthly cost).
- Two survivors scored with one sentence per score: **Wazuh 89.0 %**, **Graylog Open 69.0 %**.
- Decision record: Wazuh for the Lab 2 pilot; Graylog Open runner-up; review date October 16; rollback = snapshot of MON-01 before installation. Nothing installed yet — the choice is on paper, verifiable and dated.
- **Deliverables:** vdp_tool_selection_matrix.csv, the decision record (Table 3.2.C), the results chart (Figure 3.2.2).

Result

Wazuh 89.0 % — pilot in Lab 2

Graylog Open 69.0 % — runner-up

Removed Security Onion (K3), Zabbix (K1), Wireshark (K1, K4), cloud SIEM (K2)

Recorded September 28, 2026

Review October 16, 2026 — Lab 3, Lesson 6.5

Rollback snapshot of MON-01 before installation



Program 3.2 — class8_tool_selection.py (Python 3, no external modules)

1

Need and constraints. NEED is the one-sentence need N-01; CONSTRAINTS is the list of 3.2.2; the program prints both first — the order of the procedure is part of the method.

2

Knock-outs as functions. Each entry of KNOCKOUTS is a code, a sentence and a small function (lambda) returning True or False: K1 category, K2 licence cost, K3 minimum memory ≤ 8 GB, K4 alerting.

3

Criteria and candidates. CRITERIA holds seven weights summing to 100; each candidate holds four facts and a scores dictionary (0 to 5 per criterion).

4

Filter, score, rank. apply_knockouts() prints each verdict and keeps the survivors; weighted_total() computes $\sum \text{weight} \times \text{score} \div 5$; rank() sorts and prints the contribution of each criterion.

5

Record and spreadsheet. decision_record() prints the seven sections (pilot plan, review date); write_matrix() writes one row per candidate with verdicts, scores and the total or "not scored".

6

In the lecture page: the simulator lights the knock-out board green or red, then grows the bars criterion by criterion, then shows the ranking and the record — pause on any score to discuss its justification.



Figure 3.2.3 — Knock-out board and weighted totals

Figure 3.2.3 — Knock-out board (step 4) and weighted totals (step 5)

N-01: security logs of FW-01, SRV-FILE-01, PC-ACCT-01, SW-01 · 90 days / 200 GB · e-mail alert ≤ 5 min · lab VM 8 GB

	K1	K2	K3	K4	verdict
Wazuh (C7)	●	●	●	●	short-listed
Graylog Open (C1)	●	●	●	●	short-listed
Security Onion (C7)	●	●	●	●	out: K3
Zabbix (C3)	●	●	●	●	out: K1
Wireshark (C2)	●	●	●	●	out: K1, K4
Cloud SIEM (commercial) (C7)	●	●	●	●	out: K2

Weighted totals of the short list (segments = weight × score ÷ 5)



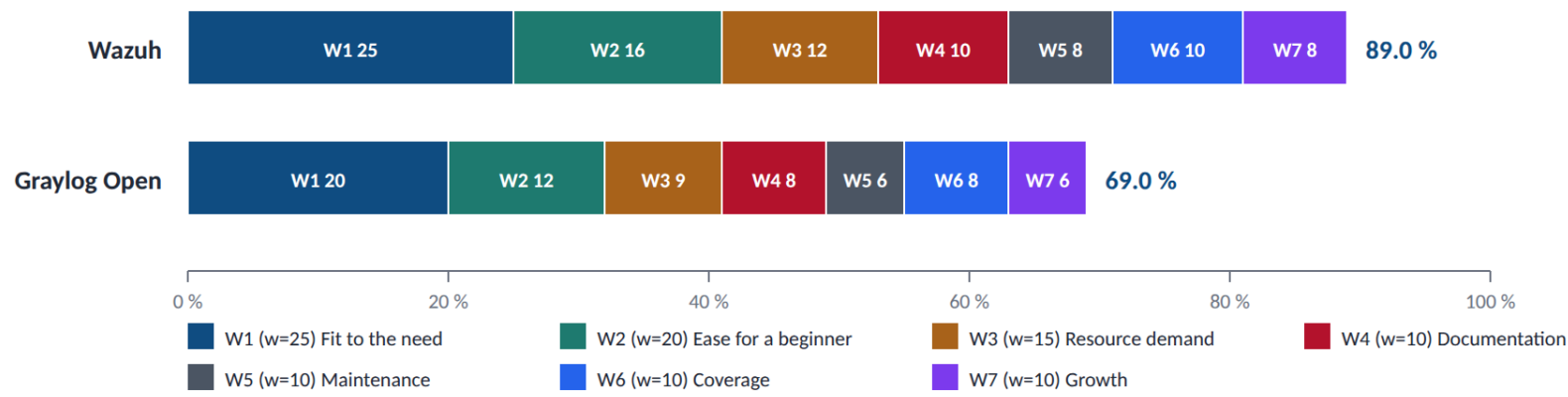
Decision: pilot Wazuh in Lab 2 (Oct 3) — runner-up Graylog Open
Pilot: snapshot first · measure install time, RAM, disk/day · 5 behaviours must alert ≤ 5 min · review Oct 16, 2026

Final state of the simulator: four lamps per candidate; two survivors scored; the decision box.



Figure 3.2.2 — Weighted totals of the short list

Figure 3.2.2 - Weighted totals of the short-listed candidates (weights sum to 100; total = sum of weight x score / 5)



Removed by the knock-out filter (not scored):

- Security Onion - K3: documented minimum RAM above the 8 GB lab VM
- Zabbix - K1: category C3 (availability), not C1/C7
- Wireshark - K1 and K4: packet analyzer, no central store, no alerting
- Cloud SIEM (commercial) - K2: monthly licence cost

Cyber.SoHo Educational Hub - 420-1C1-DW - Class 8 - Figure 3.2.2

Wazuh 89.0 % · Graylog Open 69.0 % — the difference comes mainly from fit to the need (W1) and ease for a beginner (W2).

Each segment is weight × score ÷ 5; the totals are percentages because the weights sum to 100. Generated by Program 3.2.



Table 3.2.B — Decision matrix for need N-01 (output of Program 3.2)

Candidate	Category	K1	K2	K3	K4	W1 (w=25)	W2 (w=20)	W3 (w=15)	W4 (w=10)	W5 (w=10)	W6 (w=10)	W7 (w=10)	Weighted total
Wazuh	C7	pass	pass	pass	pass	5	4	4	5	4	5	4	89.0
Graylog Open	C1	pass	pass	pass	pass	4	3	3	4	3	4	3	69.0
Security Onion	C7	pass	pass	FAIL	pass	5	2	1	4	2	4	5	not scored
Zabbix	C3	FAIL	pass	pass	pass	1	3	5	5	4	3	2	not scored
Wireshark	C2	FAIL	pass	pass	FAIL	0	3	5	5	3	2	1	not scored
Cloud SIEM (commercial)	C7	pass	FAIL	pass	pass	5	4	5	4	5	5	5	not scored

'not scored' = removed by a knock-out. Section 2 of the Lab 2 Tool Selection Record; CSV in the lecture page.



Table 3.2.C — Decision record for need N-01

Section	Content
1. Need	N-01 — collect the security event logs of FW-01, SRV-FILE-01, PC-ACCT-01 and SW-01; keep 90 days within 200 GB; e-mail alert within 5 minutes on a predefined suspicious behaviour
2. Constraints	Zero licence budget; one technician, 60 min/day; lab VM 8 GB RAM, 200 GB disk; Windows + Linux + network syslog + cloud audit export; CNC-PC-01 agentless; lab only, fictional data (Legal Notice v2.0, ss. 6.4, 7.2)
3. Long list	Wazuh; Graylog Open; Security Onion; Zabbix; Wireshark; a commercial cloud SIEM subscription
4. Knock-outs	Security Onion — K3 (documented minimum RAM above the lab VM); Zabbix — K1 (availability category, not log management); Wireshark — K1 and K4 (packet analyzer, no central store, no alerting); Cloud SIEM — K2 (monthly licence cost)
5. Scores	Wazuh 89.0 % (W1 25, W2 16, W3 12, W4 10, W5 8, W6 10, W7 8); Graylog Open 69.0 % (W1 20, W2 12, W3 9, W4 8, W5 6, W6 8, W7 6)
6. Pilot	Lab 2, Saturday, October 3, 2026: install Wazuh in MON-01 from the publisher's official site (checksum verified); measure installation time, RAM and disk per day; verify that the five predefined suspicious behaviours raise an e-mail alert within 5 minutes; snapshot before, rollback ready
7. Decision	Wazuh selected for the pilot; runner-up Graylog Open if the pilot fails on W2 or W3. Recorded on September 28, 2026 by the technician; review date October 16, 2026 (Lab 3, Lesson 6.5)

Product names for identification only (Legal Notice, section 5). A template — Lab 2 requires your own record for your own variation.



Flash cards — Lesson 3.2

1 Need before tool

A tool is selected to satisfy a stated, testable need with six elements: asset, evidence, behaviour to detect, acceptable delay, recipient of the alert, retention of the data.

2 Knock-out (must-have) criterion

A condition a candidate must satisfy to be scored at all; one failure removes it from the short list whatever its other qualities. The reason is written down. Examples: right category, zero licence cost, fits the lab VM, can alert.

3 Weighted total

Sum over the criteria of (weight × score), divided by the maximum score. With weights summing to 100 and scores from 0 to 5, the total is a percentage. Weights are fixed before any score is given.

4 Pilot

A time-boxed installation in the Authorized Lab Environment that measures what the need says: installation time, memory and disk per day, and whether each predefined suspicious behaviour raises an alert within the stated delay. Snapshot before, rollback ready.

5 Decision record

One page: need, constraints, candidates, knock-outs and reasons, scores, choice, alternatives rejected and why, date, review date. It is the starting point of Lesson 6.5 and of Lab 3.

Questions — Lesson 3.2 (1/2)

1

In a weighted decision matrix, what is a knock-out (must-have) criterion? (MCQ)

- a) The criterion with the highest weight
- b) A condition a candidate must satisfy to be scored at all; failing it removes the candidate from the short list
- c) A criterion scored from 0 to 5 like the others
- d) The cheapest candidate

2

A technician chooses a platform whose documented minimum is 16 GB of RAM for a lab virtual machine that has 8 GB. Which selection error is this? (MCQ)

- a) Choosing by popularity
- b) Ignoring licence terms
- c) Resource mismatch — a hardware constraint that should have removed the candidate at the knock-out step
- d) Forgetting the review date

3

The need is: 'show me which internal computer sends the most data to the Internet each day.' Which evidence and category serve it best? (MCQ)

- a) Full packet capture on every workstation (C2, packets)
- b) Flow records from a collector, or the connection records of FW-01 (C2 flows or C1 events)
- c) A vulnerability scanner (C6)
- d) File integrity monitoring (C5)

Answers and the explanation of every option are in the speaker notes and in the lecture page (Check / Reveal buttons).



Questions — Lesson 3.2 (2/2)

4

What is the purpose of the pilot (step 6)? (MCQ)

- a) To train the users of the company
- b) To verify in the Authorized Lab Environment, for a fixed time and with a snapshot, that the tool installs within the constraints and detects the predefined suspicious behaviours within the stated delay
- c) To negotiate the price with the vendor
- d) To replace the decision record

5

Which items belong in a decision record? (select 3) (MSQ – select 3)

- a) The need statement with its six elements
- b) The alternatives considered and the reason each was rejected
- c) The technician's preferred dashboard colour
- d) The review date
- e) The administrator password of the lab virtual machine
- f) A screenshot of a social media post recommending the tool

6

Which constraints of Val-des-Pins act directly on the selection of the Lab 2 tool? (select 3) (MSQ – select 3)

- a) Zero licence budget
- b) One part-time technician with 60 minutes per day for monitoring
- c) The company's revenue last year
- d) No agent may be installed on CNC-PC-01 (vendor contract)
- e) The brand of the office printer
- f) The technician's preference for a dark-themed dashboard

Answers and the explanation of every option are in the speaker notes and in the lecture page (Check / Reveal buttons).



Recap — what to remember

- A monitoring tool **collects, stores, and evaluates or presents** observations; it is built from agents, sensors, collectors and consoles.
- Five kinds of evidence — **events, metrics, flow records, packets, states** — give C1 log management, C2 traffic analysis, C3 availability and performance, C5 endpoint, C6 assessment; C4 intrusion detection examines the evidence for attack patterns; C7 SIEM correlates all of it.
- Six axes classify any tool: agent-based/agentless, passive/active, continuous/scheduled, signature/baseline, open-source/commercial, on-premises/cloud.
- A need is **covered** when at least one category observes the evidence it requires; the coverage map shows assets against categories; one machine can host several categories.
- Selection follows seven steps — need, constraints, long list, knock-out filter, weighted scoring, pilot, decision record — and the **weights are fixed before any score**.
- Every tool named today is lawful in the lab on Designated Targets and an offence elsewhere — section 7.2.

Numbers to remember

- 3** functions: collect, store, evaluate/present
- 4** components: agent, sensor, collector, console
- 5** kinds of evidence
- 7** categories C1–C7
- 6** classification axes
- 7** selection steps; weights sum to 100



Next steps

- **Now:** Quiz 2 (5 %), all material to date — separate document.
- **Monday, September 28 (Class 9):** full review of Topics 1 to 3 with practice questions; bring your questions on today's lessons.
- **Thursday, October 1 (Class 10, in class):** Midterm Exam, 25 %, Topics 1 to 3 up to the selection of tools.
- **Friday, October 2 (Class 11):** 3.3 installing monitoring tools and applying updates; 3.4 creating the reference database.
- **Saturday, October 3 (Class 12):** 3.5 configuring alerts; **Lab 2** — install and configure a monitoring tool on your Part 1 network, build the baseline, set the alerts.
- Bring to Lab 2: your VirtualBox lab with a fresh snapshot, your Lab 1 map and plan, and today's three deliverables adapted to your variation — coverage map, decision matrix, decision record.
- **After class:** watch the four videos (43 minutes). **Lab 1** remains due Sunday, October 11, 23:59 EDT.

Dates

Today Quiz 2 (5 %)

Mon Sept 28 Class 9 — review

Thu Oct 1 Midterm (25 %), in class

Fri Oct 2 Class 11 — 3.3, 3.4

Sat Oct 3 Class 12 — 3.5, Lab 2

Sun Oct 11 Lab 1 due, 23:59 EDT



All links used in this lecture (1/2)

Course, legal framework and safety checks

[Cyber.SoHo — Legal Notice, Terms of Use and Acceptable Use Policy, v2.0 \(LEA.D8\)](#)

· [VirusTotal check](#)

[Dawson College](#) · [VirusTotal check](#)

[Justice Laws Website \(Government of Canada\)](#) · [VirusTotal check](#)

[LégisQuébec](#) · [VirusTotal check](#)

[VirusTotal — URL scanning](#) · [VirusTotal check](#)

[Google Safe Browsing — site status](#) · [VirusTotal check](#)

Standards and guidance

[Canadian Centre for Cyber Security — Network security logging and monitoring \(ITSAP.80.085\)](#)

· [VirusTotal check](#)

[NIST SP 800-94 — Guide to Intrusion Detection and Prevention Systems](#) · [VirusTotal check](#)

[NIST SP 800-92 — Guide to Computer Security Log Management](#) · [VirusTotal check](#)

[CIS Critical Security Controls](#) · [VirusTotal check](#)

[MITRE ATT&CK — Data Sources](#) · [VirusTotal check](#)

[IETF RFC 5424 — The Syslog Protocol](#) · [VirusTotal check](#)

[IETF RFC 7011 — IPFIX](#) · [VirusTotal check](#)

Before you open any link: read the domain letter by letter and scan the address (Legal Notice, section 14.2 and Annex C). The full list with a short explanation of each link is in the lecture page.



All links used in this lecture (2/2)

Tools named in Lesson 3.1 (official sites; identification only)

[Wireshark](#) · [VirusTotal check](#)

[tcpdump and libpcap](#) · [VirusTotal check](#)

[ntop — ntopng](#) · [VirusTotal check](#)

[Graylog](#) · [VirusTotal check](#)

[OpenSearch](#) · [VirusTotal check](#)

[Elastic](#) · [VirusTotal check](#)

[Zabbix](#) · [VirusTotal check](#)

[Nagios Core](#) · [VirusTotal check](#)

[LibreNMS](#) · [VirusTotal check](#)

[Prometheus](#) · [VirusTotal check](#)

[Grafana](#) · [VirusTotal check](#)

[Suricata](#) · [VirusTotal check](#)

[Snort](#) · [VirusTotal check](#)

[Zeek](#) · [VirusTotal check](#)

[Wazuh](#) · [VirusTotal check](#)

[Security Onion](#) · [VirusTotal check](#)

[Microsoft Sysinternals — Sysmon](#) · [VirusTotal check](#)

Before you open any link: read the domain letter by letter and scan the address (Legal Notice, section 14.2 and Annex C). The full list with a short explanation of each link is in the lecture page.

[Greenbone \(OpenVAS\)](#) · [VirusTotal check](#)

[Oracle VM VirtualBox](#) · [VirusTotal check](#)

Videos and publisher indexes

[Logs and Monitoring – CompTIA Network+ N10-009 – 3.2 \(Professor Messer, 13:06\)](#) · [VirusTotal check](#)

[Security Tools – CompTIA Security+ SY0-701 – 4.4 \(Professor Messer, 14:06\)](#) · [VirusTotal check](#)

[Network Solutions – CompTIA Network+ N10-009 – 3.2 \(Professor Messer, 6:57\)](#) · [VirusTotal check](#)

[SNMP – CompTIA Network+ N10-009 – 3.2 \(Professor Messer, 8:58\)](#) · [VirusTotal check](#)

[Professor Messer — N10-009 course index](#) · [VirusTotal check](#)

[Professor Messer — SY0-701 course index](#) · [VirusTotal check](#)



Quiz 2 now — then Monday's review

Quiz 2 (5 %) covers all material to date and is provided separately.

Class 9, Monday, September 28: review of Topics 1 to 3, practice questions.

Midterm Exam, Thursday, October 1, in class (25 %): Topics 1 to 3, up to the selection of tools.

Lab 1 is due Sunday, October 11, 23:59 EDT (Léa DropBox).



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<https://cyber-soho.github.io/prevmon/00.01.%20cyber-soho%20-%20legal.html>